

60. RAM

This MCU has three types of high-speed static RAM (RAM, expansion RAM, and ECCRAM).

The RAM and expansion RAM do not have error correction but do have 1-bit parity checking.

The ECCRAM has 1-bit error correction and 2-bit error detection.

60.1 Overview

Table 60.1 lists the specifications of the RAM, expansion RAM and ECCRAM.

Table 60.1 Specifications of the RAM, Expansion RAM and ECCRAM

Item	RAM	Expansion RAM	ECCRAM
Capacity	512 Kbytes	512 Kbytes	32 Kbytes
Address Range	0000 0000h to 0007 FFFFh	0080 0000h to 0087 FFFFh	00FF 8000h to 00FF FFFFh
Memory Bus	Memory bus 1	Memory bus 3	Memory bus 3
Access	- Enabling or disabling of the RAM is selectable.*1 - Access takes one cycle whether for reading or writing.*2	- Enabling or disabling of the expansion RAM is selectable.*1 [When MEMWAIT*3 is 0] - Access takes one cycle whether for reading or writing.*2 [When MEMWAIT*3 is 1] - Access takes two cycles whether for reading or writing.*2	- Enabling or disabling of the ECCRAM is selectable.*1 [When MEMWAIT*3 is 0] - With the ECC function disabled: Access takes two cycles whether for reading or writing.*2 - With the ECC function enabled (and there are no errors): Access takes two cycles whether for reading or writing.*2 - With the ECC function enabled (and an error is found): Access takes three cycles whether for reading or writing.*2 [When MEMWAIT*3 is 1] - With the ECC function disabled: Access takes three cycles whether for reading or writing.*2 - With the ECC function enabled (and there are no errors): Reading takes three cycles and writing takes four cycles.*2 - With the ECC function enabled (and an error is found): Access takes five cycles whether for reading or writing.*2
Data Retention	Data retention is not available on deep software standby.		
Low Power Consumption	The RAM, expansion RAM, and ECCRAM are independently transferrable to the module stop state.		
Error Checking	- Parity checking: 1-bit error detection - When an error is found, a non-maskable interrupt or an interrupt is generated.		- ECC error correction 1-bit error correction and 2-bit error detection - When an error is found, a non-maskable interrupt or an interrupt is generated.

Note 1. This is selectable for the RAM and expansion RAM by the SYSCR1.RAME bit, and for the ECCRAM by the SYSCR1.ECCRAM bit. For details on SYSCR1, refer to section 3.2.3, System Control Register 1 (SYSCR1).

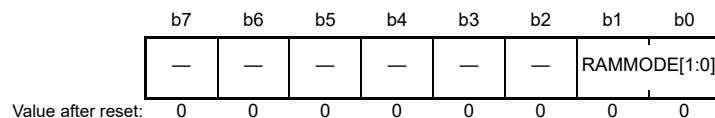
Note 2. When accessing across the 8-byte boundary, the number of cycles is doubled.

Note 3. If the frequency of ICLK is above 120 MHz, the setting of the MEMWAIT register must be changed. For details, refer to section 9.2.2, Memory Wait Cycle Setting Register (MEMWAIT).

60.2 Register Descriptions

60.2.1 RAM Operating Mode Control Register (RAMMODE)

Address(es): 0008 1200h

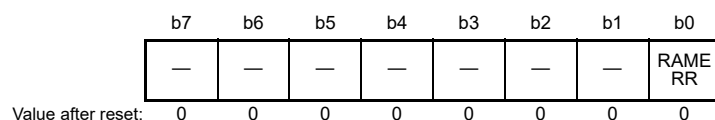


Bit	Symbol	Bit Name	Description	R/W
b1, b0	RAMMODE[1:0]	RAM Operating Mode Select	b1 b0 0 0: Parity checking is disabled. 0 1: Parity checking is enabled. Settings other than above are prohibited.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R

The RAMMODE register is write-protected by the RAM protection register (RAMPRCR). Before writing to the RAMMODE register, set the RAMPRCR.RAMPRCR bit to 1 to enable writing to it. Set the RAMMODE register before starting access to the RAM. If this register is modified after accessing to the RAM, RAM operation is not guaranteed.

60.2.2 RAM Error Status Register (RAMSTS)

Address(es): 0008 1201h



Bit	Symbol	Bit Name	Description	R/W
b0	RAMERR	RAM Error Status Flag	0: A parity check error has not occurred. 1: A parity check error has occurred.	R/(W)*1
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. Only 0 can be written to clear the flag.

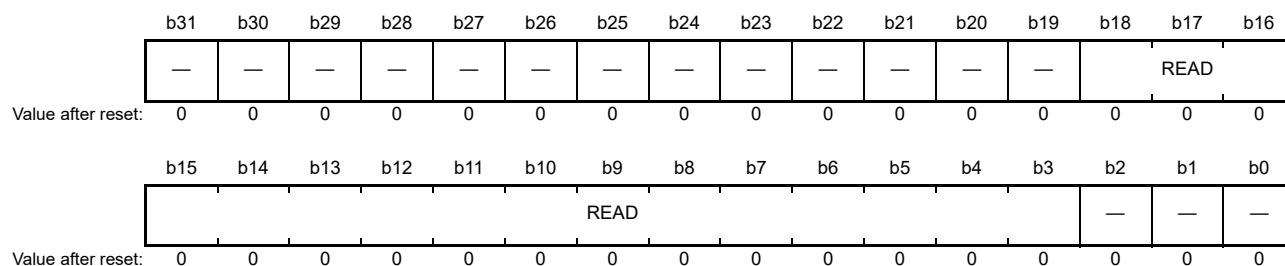
When parity checking is enabled, the RAMERR flag is set to 1 if a parity check error is detected. The RAM error interrupt request is also generated at this time.

When parity checking is disabled, the RAMERR flag is not set to 1 because no parity check error is detected.

Writing 0 to the RAMERR flag clears the RAM error interrupt request corresponding to the parity check error.

60.2.3 RAM Error Address Capture Register (RAMECAD)

Address(es): 0008 1208h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b18 to b3	READ	Error Address	The address where an error is found is read.	R
b31 to b19	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

When parity checking is enabled, this register will hold the address where a parity check error was found.

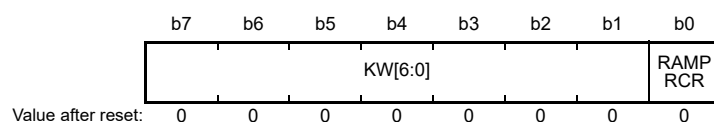
The address of the 8-byte boundary below the location where the error was found is stored in this register at the same time the RAMSTS.RAMERR flag is set to 1.

The error address is not updated when the RAMERR flag is 1 (error has occurred). Its value does not change when parity checking is disabled because no parity check error is detected.

The RAMECAD register is initialized only by a reset.

60.2.4 RAM Protection Register (RAMPRCR)

Address(es): 0008 1204h



Bit	Symbol	Bit Name	Description	R/W
b0	RAMPRC R	RAMMODE Register Write Control	0: Disables writing to the RAMMODE register. 1: Enables writing to the RAMMODE register.	R/W
b7 to b1	KW[6:0]	Write Key Word	Enables or disables the rewriting of the RAMPRCR register. When rewriting the RAMPRCR register, write 1111000b to the KW[6:0] bits.	R/W

Writing 1 to the RAMPRCR bit is possible when KW[6:0] = 1111000b. Otherwise writing to RAMPRCR clears the bit to 0. The value of KW[6:0] is read as 0000000b.

The targets for write protection by the RAMPRCR register is the RAM operating mode control register (RAMMODE). Once the RAMPRCR bit is set to 1, writing to RAMMODE register is enabled until the RAMPRCR bit is cleared to 0. Clear the RAMPRCR bit to 0 after writing to RAMMODE register.

60.2.5 Expansion RAM Operating Mode Control Register (EXRAMMODE)

Address(es): 0008 1240h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	EXRAMMODE[1:0]	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	EXRAMMODE[1:0]	Expansion RAM Operating Mode Select	b1 b0 0 0: Parity checking is disabled. 0 1: Parity checking is enabled. Settings other than above are prohibited.	R/W
b7 to b2	—	Reserved	These bits are read as 0.	R

The EXRAMMODE register is write-protected by the expansion RAM protection register (EXRAMPRCR). Before writing to the EXRAMMODE register, set the EXRAMPRCR.EXRAMPRCR bit to 1 to enable writing to it. Set the EXRAMMODE register before starting access to the expansion RAM. If this register is modified after accessing to the expansion RAM, correct operation is not guaranteed.

60.2.6 Expansion RAM Error Status Register (EXRAMSTS)

Address(es): 0008 1241h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	EXRAMERR
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	EXRAMERR	Expansion RAM Error Status Flag	0: A parity check error has not occurred. 1: A parity check error has occurred.	R/(W)*1
b7 to b1	—	Reserved	These bits are read as 0.	R

Note 1. Only 0 can be written to clear the flag.

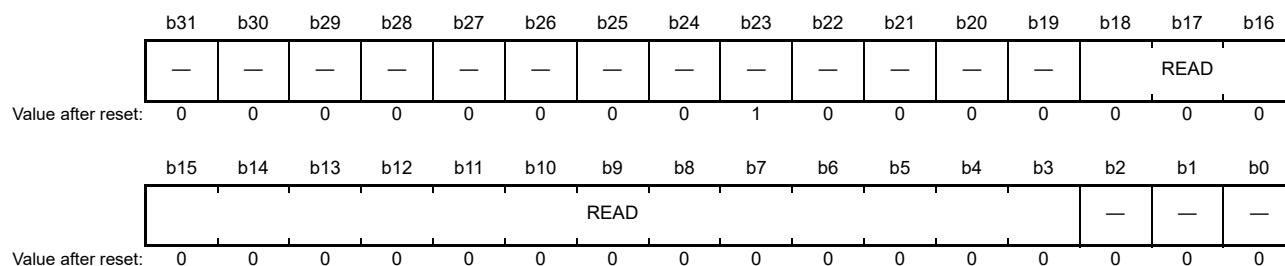
When parity checking is enabled, the EXRAMERR flag is set to 1 if a parity check error is detected. The expansion RAM error interrupt request is also generated at this time.

When parity checking is disabled, the EXRAMERR flag is not set to 1 because no parity check error is detected.

Writing 0 to the EXRAMERR flag clears the expansion RAM error interrupt request corresponding to the parity check error.

60.2.7 Expansion RAM Error Address Capture Register (EXRAMECAD)

Address(es): 0008 1248h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b18 to b3	READ	Error Address	The address where an error is found is read.	R
b22 to b19	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b23	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

When parity checking is enabled, this register will hold the address where a parity check error was found.

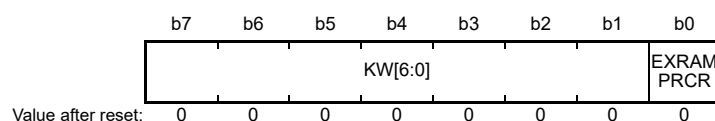
The address of the 8-byte boundary below the location where the error was found is stored in this register at the same time the EXRAMSTS.EXRAMERR flag is set to 1.

The error address is not updated when the EXRAMERR flag is 1 (error has occurred). Its value does not change when parity checking is disabled because no parity check error is detected.

The EXRAMECAD register is initialized only by a reset.

60.2.8 Expansion RAM Protection Register (EXRAMPRCR)

Address(es): 0008 1244h



Bit	Symbol	Bit Name	Description	R/W
b0	EXRAMPRCR	EXRAMMODE Register Write Control	0: Disables writing to the EXRAMMODE register. 1: Enables writing to the EXRAMMODE register.	R/W
b7 to b1	KW[6:0]	Write Key Word	Enables or disables the rewriting of the EXRAMPRCR register. When rewriting the EXRAMPRCR register, write 1111000b to the KW[6:0] bits.	R/W

Writing 1 to the EXRAMPRCR bit is possible when KW[6:0] = 1111000b. Otherwise writing to EXRAMPRCR clears the bit to 0. The value of KW[6:0] is read as 0000000b.

The targets for write protection by the EXRAMPRCR register is the expansion RAM operating mode control register (EXRAMMODE). Once the EXRAMPRCR bit is set to 1, writing to EXRAMMODE register is enabled until the EXRAMPRCR bit is cleared to 0. Clear the EXRAMPRCR bit to 0 after writing to EXRAMMODE register.

60.2.9 ECCRAM Operating Mode Control Register (ECCRAMMODE)

Address(es): 0008 12C0h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	RAMMOD[1:0]	
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b1, b0	RAMMOD[1:0]	RAM Operating Mode Select	b1 b0 0 0: ECCs are disabled. 0 1: Setting prohibited. 1 0: ECCs are enabled without error checking. 1 1: ECCs are enabled with error checking.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The ECCRAMMODE register is write-protected by the ECCRAM protection register (ECCRAMPRCR). Before writing to the ECCRAMMODE register, set the corresponding bit in the ECCRAMPRCR register to 1 to enable writing to it. Be sure to read this register immediately after writing to it, and then compare the read and written values. If this operation is not performed, access to the ECCRAM is not guaranteed. Also, do not write to this register while accessing to the ECCRAM.

60.2.10 ECCRAM 2-Bit Error Status Register (ECCRAM2STS)

Address(es): 0008 12C1h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	ECC2ERR
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	ECC2ERR	2-Bit ECC Error Status	0: A 2-bit ECC error has not occurred. 1: A 2-bit ECC error has occurred.	R/(W)*1
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only 0 can be written to clear the bit.

When the ECCRAMMODE.RAMMOD[1:0] bits are set to 11b, the ECC2ERR flag is set to 1 if a 2-bit error is found. The RAM error interrupt is also generated at this time. Writing of 0 to the ECC2ERR flag leads to clearing of the RAM error interrupt having the 2-bit ECC error as its source.

60.2.11 ECCRAM 1-Bit Error Information Update Enable Register (ECCRAM1STSEN)

Address(es): 0008 12C2h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	ECC1S TSEN
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	ECC1STS EN	1-Bit ECC Error Information Update Enable	0: Disables updating of the 1-bit ECC error information. 1: Enables updating of the 1-bit ECC error information.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The ECCRAM1STSEN register enables or disables updating of the ECCRAM 1-bit error status register (ECCRAM1STS) in response to a 1-bit error in the ECCRAM.

The ECCRAMPRCR register can protect this register against writing. Accordingly, change the effective bit in the ECCRAMPRCR register to the enabled setting before attempting to write to the ECCRAM1STSEN register.

Always be sure to read this register immediately after writing to it, and then compare the read and written values. Do not attempt access to the ECCRAM during this process. Also, do not write to this register while accessing to the ECCRAM.

60.2.12 ECCRAM 1-Bit Error Status Register (ECCRAM1STS)

Address(es): 0008 12C3h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	ECC1E RR
0	0	0	0	0	0	0	0

Value after reset:

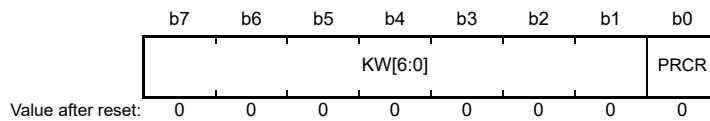
Bit	Symbol	Bit Name	Description	R/W
b0	ECC1ERR	1-Bit ECC Error Status	0: A 1-bit ECC error has not occurred. 1: A 1-bit ECC error has occurred.	R/(W)*1
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only 0 can be written to clear the bit.

The ECC1ERR bit becomes 1 when an ECC 1-bit error is detected in response to an ECC 1-bit error being encountered while the setting of the ECCRAM1STSEN.ECC1STSEN bit is 1 and the ECCRAMMODE.RAMMOD[1:0] bits are set as ECCs are enabled with error checking. The RAM error interrupt is also generated at this time. Writing of 0 to the ECC1ERR bit leads to clearing of the RAM error interrupt having the 1-bit ECC error as its source.

60.2.13 ECCRAM Protection Register (ECCRAMPRCR)

Address(es): 0008 12C4h



Bit	Symbol	Bit Name	Description	R/W
b0	PRCR	ECCRAMMODE and ECCRAM1STSEN Registers Write Control	0: Disables writing to the ECCRAMMODE and ECCRAM1STSEN registers. 1: Enables writing to the ECCRAMMODE and ECCRAM1STSEN registers.	R/W
b7 to b1	KW[6:0]	Write Key Word	Enables or disables the rewriting of the ECCRAMPRCR register. When rewriting the ECCRAMPRCR register, write 1111000b to the KW[6:0] bits.	R/W

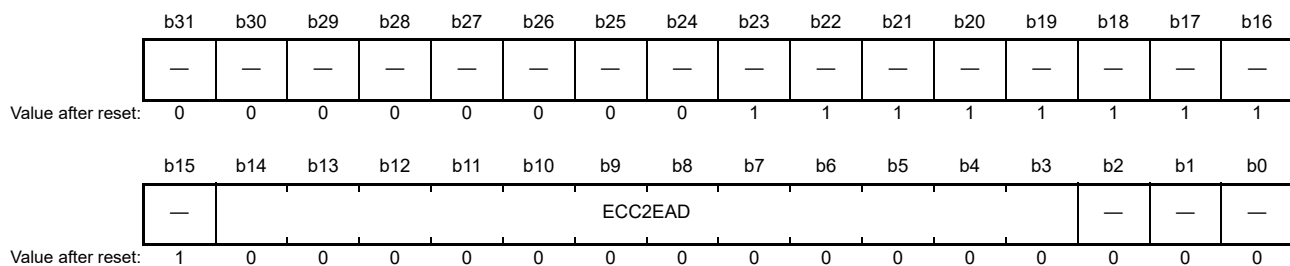
Writing 1 to the PRCR bit is possible when KW[6:0] = 1111000b.

Otherwise writing to PRCR clears the bit to 0. KW[6:0] is read as 0000000b.

The targets for write protection by the ECCRAMPRCR register are the ECCRAMMODE and ECCRAM1STSEN registers. Once the PRCR bit is set to 1, writing to ECCRAMMODE and ECCRAM1STSEN registers is enabled until the PRCR bit is cleared to 0. Clear the PRCR bit to 0 after writing to ECCRAMMODE and ECCRAM1STSEN registers.

60.2.14 ECCRAM 2-Bit Error Address Capture Register (ECCRAM2ECAD)

Address(es): 0008 12C8h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0.	R
b14 to b3	ECC2EAD	2-Bit ECC Error Address	The address where a 2-bit ECC error was found is read.	R
b23 to b15	—	Reserved	These bits are read as 1.	R
b31 to b24	—	Reserved	These bits are read as 0.	R

When the ECCRAMMODE.RAMMOD[1:0] bits are set to 11b, this register will hold the address where a 2-bit ECC error was found. Specifically, the address of the 8-byte boundary below the location where the error was found is written to the ECC2EAD bits of this register at the same time as the ECCRAM2STS.ECC2ERR flag is set. The address is not updated whenever the ECC2ERR flag is set to 1 to indicate an error. Its value does not change if ECCs are disabled. The value is initialized by a reset, and this is the only condition that leads to clearing of the register.

60.2.15 ECCRAM 1-Bit Error Address Capture Register (ECCRAM1ECAD)

Address(es): 0008 12CCh

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	ECC1EAD												—	—	—
Value after reset:	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0.	R
b14 to b3	ECC1EAD	1-Bit ECC Error Address	The address where a 1-bit ECC error was found is read.	R
b23 to b15	—	Reserved	These bits are read as 1.	R
b31 to b24	—	Reserved	These bits are read as 0.	R

When an ECC 1-bit error is found while the setting of the ECCRAM1STSEN.ECC1STSEN bit is 1 and the setting of the ECCRAMMODE.RAMMOD[1:0] bits are “ECCs are enabled with error checking”, the effective bits of this register retain the address where the error was found. Specifically, the address of the 8-byte boundary below the location where the error was found is written to the effective bits of this register at the same time as the ECCRAM1STS.ECC1ERR bit is set.

The address is not updated whenever the ECC1ERR bit is set to 1 to indicate an error. Its value does not change if ECCs are disabled. The value is initialized by a reset, and this is the only condition that leads to clearing of the register.

60.2.16 ECCRAM Protection Register 2 (ECCRAMPRCR2)

Address(es): 0008 12D0h

	b7	b6	b5	b4	b3	b2	b1	b0
	KW2[6:0]							PRCR2
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PRCR2	ECCRAMETST Register Write Control	0: Writing to ECCRAMETST is disabled. 1: Writing to ECCRAMETST is enabled.	R/W
b7 to b1	KW2[6:0]	Write Keyword	Enables or disables the rewriting of the ECCRAMPRCR2 register. When rewriting the ECCRAMPRCR2 register, write 1111000b to the KW2[6:0] bits.	R/W

When KW2[6:0] = 1111000b, writing 1 to the PRCR2 bit is enabled.

Otherwise writing to PRCR2 clears the bit to 0. The KW2[6:0] bits are read as 0000000b.

The target for write protection by the PRCR2 bit is the ECCRAM test control register (ECCRAMETST). Once the PRCR2 bit is set to 1, writing to the ECCRAMETST register is enabled until the PRCR2 bit is cleared to 0. After writing to the ECCRAMETST register, clear the PRCR2 bit to 0.

60.2.17 ECCRAM Test Control Register (ECCRAMETST)

Address(es): 0008 12D4h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	TSTBYP
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TSTBYP	ECC Bypass Select	0: ECC bypass is disabled. 1: ECC bypass is enabled.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

The protection register 2 (ECCRAMPRCR2) protects this register against writing. Accordingly, change the effective bit in the ECCRAMPRCR2 register to the enabled setting before attempting to write to the ECCRAMETST register. Do not write to ECCRAMETST while access to ECCRAM is in progress.

Eight of the 72 bits of data in the ECCRAM are used for the ECC. When the TSTBYP bit is set to 1 to select bypassing of the ECC circuit, the 8 bits for the ECC become directly accessible. Setting the TSTBYP bit for bypassing of the ECC is only possible when the ECC is disabled by setting the ECCRAMMODE.RAMMOD[1:0] bit to 00b.

60.3 Operation

60.3.1 Correction of ECC Errors

Enabling and disabling of ECC error correction can be selected through the ECCRAMMODE register setting. In the initial state, ECC error correction is disabled. The ECC used in this device is SEC-DED (Single-Error Correction and Double-Error Detection Code).

When ECCs are enabled, an 8-bit ECC code is appended to 64-bit data for writing. For reading, 72-bit (data: 64 bits, ECC code: 8 bits) data is read out from the ECCRAM.

60.3.1.1 ECC Enabled and Error Checking Selected

If a 1-bit error is found, the error is corrected, and if the setting of the ECCRAM1STSEN.ECC1STSEN bit is 1, the ECCRAM1STS.ECC1ERR bit is set to 1. When an error is detected, the address of the 8-byte boundary including the location where the error was found is set to the ECCRAM1ECAD register. If a 2-bit error occurs, error detection is done and the ECCRAM2STS.ECC2ERR flag is set to 1, though error correction is not performed. When an error is detected, the address of the 8-byte boundary including the location where the error was found is set to the ECCRAM2ECAD register. In addition, there is no way to confirm the location where the error was found within the 8-byte boundary. Therefore, when the correct data is written after the occurrence of an error, update all the data in the 8-byte boundary.

60.3.1.2 ECC Enabled and Error Checking not Selected

In reading, the error is corrected if a 1-bit error is found, but the corrected value is not written to the ECCRAM, and the ECC code is neither generated nor written back. In writing, the corrected value is written, and the ECC code is generated and written back.

In writing, write data is replaced and ECC codes are generated then written back.

Since the ECCRAM data is undefined after power on and release from deep software standby mode, accessing the ECCRAM when ECCs are enabled and error checking is selected causes an ECC error to occur. Therefore, when using ECC error correction, initial writing to the area to be used in the ECCRAM should be done beforehand when ECCs are enabled but error checking is not selected. Since the ECC error correction is done in 8-byte units, the initial write must be done for the $8 \times N$ byte area including the area to be used.

60.3.2 Parity Checking

Enabling and disabling of parity checking for the RAM or expansion RAM can be selected through the RAMMODE or EXRAMMODE register setting. In the initial state, parity checking is disabled. Even parity checking is used in this device.

1-bit parity check code is added to each 1-byte data for writing, and the parity is checked for reading.

If a 1-bit error is detected in the 1 byte when the parity is checked for reading, a RAM error interrupt can be generated. If a 2-bit error or more is detected in the 1 byte, errors cannot be correctly detected.

After power-on, parity check code is undefined until data is written. To use parity checking, write the initial value to all areas while parity checking is enabled before accessing to the RAM immediately after a reset.

Operation cannot be guaranteed if access is made to an area where the initial value is not written.

60.3.3 RAM Error Interrupt Function

With ECCs enabled and error checking selected, if any of the ECCRAM2STS.ECC2ERR bit that indicates an ECC 2-bit error, ECCRAM1STS.ECC1ERR bit that indicates an ECC 1-bit error, EXRAMSTS.EXRAMERR bit that indicates a parity error, or RAMSTS.RAMERR bit that indicates a parity error when parity checking is enabled for the normal RAM changes to 1, a RAM error interrupt is generated.

Writing 0 to a bit that has been set clears the RAM error interrupt due to the corresponding source.

If the error from the other source is still occurring at this time, the RAM error interrupt will remain generated.

When the ECC 1-bit error interrupt is to be masked, set the ECCRAM1STSEN.ECC1STSEN bit to 0 to disable updating of the ECC1ERR bit. A RAM error interrupt will not be generated while ECCs are disabled and when ECCs are enabled but error checking is not selected.

60.3.4 Interrupt Source

Interrupt sources from the RAM are the RAM error interrupt generated by the RAM or expansion RAM when a parity error is found, and the RAM error interrupt generated by the ECCRAM when an ECC error is found, both of which can be set up for use as non-maskable interrupts or normal interrupts. For details, refer to **section 15, Interrupt Controller (ICUD)**.

Table 60.2 RAM Interrupt Source

Name	Interrupt Source	DTC Activation	DMAC Activation
RAMERR	Parity check error or ECC error	Not possible	Not possible

60.3.5 ECC Decoder Testing

In this MCU, the user can test the ECC code control circuit for the ECCRAM.

Figure 60.1 shows the flow for testing of the ECC code control circuit.

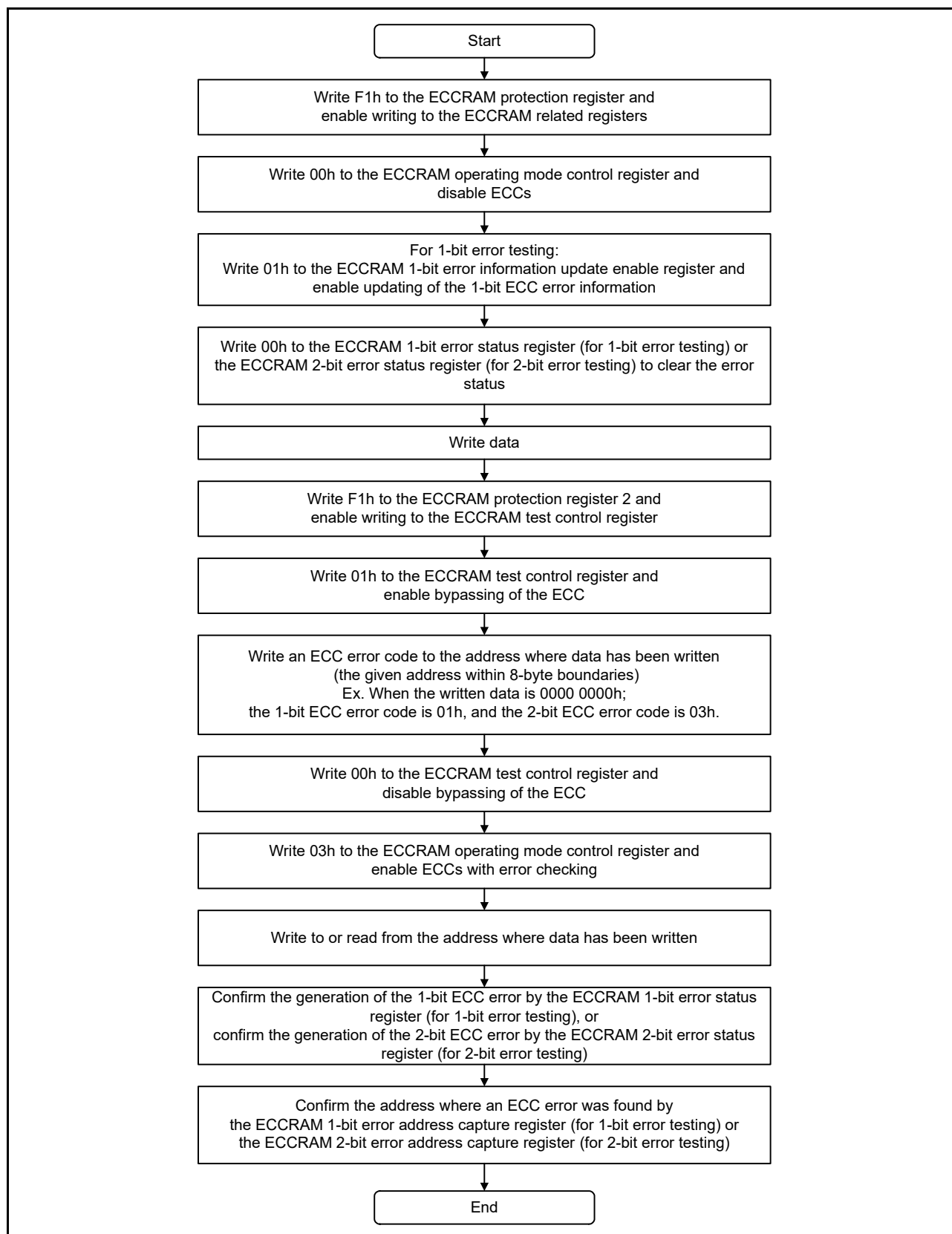


Figure 60.1 Flow for Testing of the ECC Code Control Circuit

60.4 Usage Notes

60.4.1 Low Power Consumption Function

Power consumption can be reduced by setting module stop control register C (MSTPCRC) to stop supply of the clock signal to the RAM, expansion RAM and ECCRAM.

Setting the MSTPCRC.MSTPC0 bit to 1 stops supply of the clock signal to the RAM. Setting the MSTPCRC.MSTPC2 bit to 1, stops supply of the clock signal to the expansion RAM. Setting the MSTPCRC.MSTPC6 bit to 1 stops supply of the clock signal to the ECCRAM.

The RAM, expansion RAM, and ECCRAM can be individually placed in the module stop state, in which the supply of the clock signal to the given module is stopped.

The RAM, expansion RAM, and ECCRAM operate after a reset.

The RAM, expansion RAM, and ECCRAM are not accessible in the module stop state.

Do not allow transitions to the module stop state while accessing to the RAM, expansion RAM, or ECCRAM.

Access to the RAM, expansion RAM, and ECCRAM in the module stop state is prohibited. If access is attempted, correct operation is not guaranteed.

For details on the MSTPCRC register, refer to section 11, Low Power Consumption.

60.4.2 Notes on Using Error Checking of RAM, Expansion RAM and ECCRAM

Data in RAM, ECCRAM, and expansion RAM are undefined when the power is turned on. Therefore, parity check errors or ECC errors occur if the data are read before initialization. The RAM, expansion RAM, and ECCRAM are read in 8-byte (64-bit) units. Initialize them on 8-byte boundaries.

When a program is executed in the RAM, expansion RAM, or ECCRAM with the parity check or the ECC function enabled, initialize the RAMs in consideration of possible instruction prefetching by the CPU. Instruction prefetching can be performed up to 32 bytes. The initialization must thus cover extra 24 to 31 bytes from the last address of the program.

60.4.3 Notes on Self-Diagnosis of the RAM

A write buffer is mounted for the RAM*¹. When the same address is read after a write operation, data in the write buffer, rather than in the memory cell of the RAM may be read. When the RAM is self-diagnosed, confirm that the data have been written by following the procedure below so that data will not be read from the write buffer.

- (1) Write data to the address targeted for diagnosis.
- (2) Write data to an address which is at least 4 addresses away from the that in (1).
- (3) Read the data from the address in (1).

Note 1. This does not apply to the expansion RAM and ECCRAM.